CPGSV17 Blocked-Basis Chi Families and Uniform BNT Labels

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1 Source statement

Cirac–Pérez-García–Schuch–Verstraete 2017 state in [CPGSV17, Theorem IV.13(ii)] that there is a set of diagonal matrices

$$\chi_{\alpha,\beta,\gamma}$$

with positive entries such that, for every length L, the operators $O_L(M_\alpha)$ linearly span an algebra whose structure coefficients satisfy

$$c^{(L)}_{\alpha,\beta,\gamma} = \operatorname{tr}(\chi^L_{\alpha,\beta,\gamma}).$$

The important point for the present note is that the labels α, β, γ are BNT labels, not basis labels depending on L. Thus the same diagonal matrix $\chi_{\alpha,\beta,\gamma}$ is used for all positive lengths; only the exponent changes.

Here $O_L(M_\alpha)$ denotes the length-L operator associated to the normal tensor labelled by α in the basis of normal tensors. The phrase "BNT labels" refers to these labels α, β, γ. The support algebra $\mathcal{A}_n$ below is the formal support algebra chosen for the blocked length n in the Lean development.

2 Formalized blocked-basis analogue

The current blocked coefficient construction in `TNLean/MPS/MPDO/AlgebraStructure.lean` chooses bases of the support algebras $\mathcal{A}_n$. Its multiplication coefficients have indices

$$i, j \in \operatorname{Basis}(\mathcal{A}_n), \qquad k \in \operatorname{Basis}(\mathcal{A}_{2n}),$$

because multiplication is represented as

$$\mathcal{A}_n \times \mathcal{A}_n \longrightarrow \mathcal{A}_{2n}.$$

The declarations MPOTensor.AlgebraStructureData.BlockedStructureChiFamily and MPOTensor.AlgebraStructureData.HasBlockedStructureChiTracePowerForm therefore express a positive-length, length-dependent blocked-basis analogue:

$$c_{i,j,k}^{(n)} = \sum_r \chi_{n,i,j,k,r}^n = \mathrm{tr}(\chi_{n,i,j,k}^n).$$

Here n is the source blocking length of the two factors in $\mathcal{A}_n$, while the coefficient k is taken in the basis of $\mathcal{A}_{2n}$. The predicate is intentionally restricted to $0 < n$, avoiding the degenerate length-zero constraint $c_{i,j,k}^{(0)} = \dim(\chi_{0,i,j,k})$, which is not part of the intended positive-chain statement.

3 Gap

Verdict: scope restriction.

This blocked-basis analogue is not the uniform BNT-label statement of Theorem IV.13(ii). It allows the diagonal matrix to depend on n, because the available indices themselves depend on the blocked bases $\mathcal{A}_n$ and $\mathcal{A}_{2n}$. The source theorem instead provides one matrix for each triple of BNT labels and uses that same matrix for every length L.

There is a second scope restriction in the operation being compared. In the paper, the coefficients $c_{\alpha,\beta,\gamma}^{(L)}$ belong to the same-length operator algebra:

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma c_{\alpha,\beta,\gamma}^{(L)} O_L(M_\gamma).$$

The current Lean coefficient family comes from the blocked support-algebra product

$$\mathcal{A}_n \times \mathcal{A}_n \longrightarrow \mathcal{A}_{2n},$$

so the two input factors have source length n, while the result is expanded in a basis at length $2n$. The exponent n in

$$c_{i,j,k}^{(n)} = \mathrm{tr}(\chi_{n,i,j,k}^n)$$

is therefore the source length of the two factors, not the codomain length. This convention is only a blocked-basis analogue; it is not a replacement for the same-length multiplication statement in Theorem IV.13(ii).

This note isolates the chi-uniformity and same-length coefficient comparison part of Theorem IV.13(ii). The idempotent condition

$$m_\gamma = \sum_{\alpha,\beta} c_{\alpha,\beta,\gamma}^{(1)} m_\alpha m_\beta$$

from paper lines 981–985 is a separate condition in the same theorem; it is not represented by the blocked-basis trace-power predicate.

Consequently, MPOTensor.AlgebraStructureData.HasBlockedStructureChiTracePowerForm should not be cited as a formalization of Theorem IV.13(ii). It is a local blocked-basis coefficient statement, useful only after the uniform BNT-label statement has been separated from the chosen blocked bases.

4 Relation to the algebra-to-fusion converse

Let $E = E_1$ be the one-site transfer map. The present algebra predicate gives the fixed-point-space identity

$$\mathrm{Fix}\big((E^n)^\dagger\big) = \mathrm{Fix}(E^\dagger), \qquad n > 0.$$

This rules out root-of-unity peripheral eigenvalues of $E^\dagger$, but it does not imply the fusion-side idempotence equation $E^2 = E$.

The converse direction (ii) $\Rightarrow$ (i) in the proof of Theorem IV.13 uses a stronger coefficient argument. In Appendix C.4, lines 2046–2085, the authors compare the two BNT expansions coming from the one-site and two-site decompositions. The source identity `eq:algebra-appendix` is stated at lines 1933–1936 and used in this converse at lines 2048–2050:

$$\sum_\gamma \mathrm{tr}(\nu_\gamma^L) O_L(A_\gamma) = \sum_{\alpha,\beta,\gamma} \mathrm{tr}\big((\mu_\alpha \otimes \mu_\beta \otimes \chi_{\alpha,\beta,\gamma})^L\big) O_L(M_\gamma).$$

Since both A_γ and M_γ are BNT, the comparison first gives, up to relabelling (line 2053),

$$A_\gamma = e^{i\theta_\gamma} Z_\gamma M_\gamma Z_\gamma^{-1}.$$

Proposition IV.12 then removes the phase and identifies Z_γ with a unitary U_γ, as in line 2057, so

$$A_\gamma = U_\gamma M_\gamma U_\gamma^{-1}$$

with U_γ unitary. Since, for all sufficiently large L, the operators $O_L(A_\gamma) = O_L(M_\gamma)$ are linearly independent, and all coefficients are positive, the simple-matrix lemma (lines 1155–1163, used at line 2058) gives the multiset identity

$$\{\nu_{\gamma,k}\}_k = \{\mu_{\alpha,k_1}\mu_{\beta,k_2}\chi_{\alpha,\beta,\gamma,k_3}\}_{\alpha,\beta,k_1,k_2,k_3}.$$

Summing this identity gives

$$\mathrm{tr}(\nu_\gamma) = \sum_{\alpha,\beta} m_\alpha m_\beta c^{(1)}_{\alpha,\beta,\gamma} = m_\gamma.$$

Only after this trace equality is obtained do the normalizing maps in [CPGSV17, lines 2065–2085] have the form

$$\widetilde{R}_2\Big(\bigoplus_\gamma X_\gamma\Big) = \bigoplus_\gamma \frac{\nu_\gamma}{m_\gamma} \otimes X_\gamma, \qquad \widetilde{R}_1\Big(\bigoplus_\gamma X_\gamma\Big) = \bigoplus_\gamma \frac{\mu_\gamma}{m_\gamma} \otimes X_\gamma.$$

These are the maps used in $T = \widetilde{R}_2\widetilde{T}R_1$ and $S = \widetilde{R}_1\widetilde{S}R_2$. The first denominator is normalized by $\mathrm{tr}(\nu_\gamma) = m_\gamma$, proved above; the second by the defining equality $m_\gamma = \mathrm{tr}(\mu_\gamma)$ for the original block coefficient.

The present algebra predicate MPOTensor.IsRFP˙MPDO˙via˙algebra does not include these BNT-label coefficients, the positive $\chi_{\alpha,\beta,\gamma}$, or the length-one idempotent trace identity. It proves the fixed-point-space equation above, not the displayed BNT coefficient comparison. Thus the source-faithful route to the fusion formulation must construct the BNT-label theorem witness from the MPDO tensor and then use the one-site retract criterion MPOTensor.fusionIsometryData˙one˙iff˙isRFP.

5 BNT-label coefficient objects and remaining elimination plan

The same-length coefficient family $c^{(L)}_{\alpha,\beta,\gamma}$, indexed by fixed BNT labels, is represented separately from the blocked-basis coefficients.[1] When the Appendix C.4 construction has already produced a diagonal $\chi_{\alpha,\beta,\gamma}$-family, there is also a canonical coefficient family

$$c^{(L)}_{\alpha,\beta,\gamma} = \sum_r \chi^L_{\alpha,\beta,\gamma,r}.$$

This is recorded separately so that the χ-family can serve as the single source of those coefficients in the source-side special case.[2] A same-length BNT product predicate records the algebra identity

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma c^{(L)}_{\alpha,\beta,\gamma} O_L(M_\gamma)$$

for an abstract family of length-L BNT operators.[3] The idempotent condition

$$m_\gamma = \sum_{\alpha,\beta} c^{(1)}_{\alpha,\beta,\gamma} m_\alpha m_\beta$$

is also represented as a predicate on the trace scalars $m_\alpha = \mathrm{tr}(\mu_\alpha)$.[4] A positive BNT-label χ-witness consists of a length-independent diagonal family $\chi_{\alpha,\beta,\gamma}$, positivity of every diagonal entry, and the positive-length trace-power identity.[5] If the coefficients are chosen canonically from the same χ-family, then positivity of the diagonal entries alone gives such a witness.[6] A blocked-basis comparison predicate records that the chosen blocked multiplication coefficients are the pullback of the source-length BNT-label coefficients along a source label map from $\mathcal{A}_n$ and a target label map from $\mathcal{A}_{2n}$. The label-assignment datum is separated from the coefficient equality, so the Appendix C.3 construction target can be stated without also assuming the comparison formula.[7] This predicate inherits the second scope restriction above: its left hand side is expanded in a basis of $\mathcal{A}_{2n}$, while the source theorem's BNT product law is same-length. It therefore records the formal blocked-basis comparison predicate; constructing the source label maps from an MPDO tensor is still part of the Appendix C.3–C.4 witness construction. The predicate is not itself the same-length product

[1] MPOTensor.BNTLabelCoefficientFamily.

[2] MPOTensor.BNTLabelCoefficientFamily.ofChi, MPOTensor.BNTLabelCoefficientFamily.ofChi_coeff, MPOTensor.BNTLabelCoefficientFamily.ofChi_coeff_eq_trace_matrix_pow, MPOTensor.BNTLabelCoefficientFamily.ofChi_hasPositiveLengthChiTracePowerForm.

[3] MPOTensor.BNTLabelOperatorFamily.HasSameLengthProductForm.

[4] MPOTensor.BNTLabelTraceScalarFamily.HasIdempotentCoefficientForm.

[5] MPOTensor.PositiveBNTLabelChiTracePowerForm.

[6] MPOTensor.PositiveBNTLabelChiTracePowerForm.ofChi.

[7] MPOTensor.BNTBlockedBasisLabelAssignment, MPOTensor.BNTBlockedBasisLabelAssignment.blockedLabel, MPOTensor.BNTBlockedBasisCoefficientComparison.

statement of Theorem IV.13(ii). These declarations specify the BNT-label coefficient objects and the comparison statement, but they do not yet construct the operators, trace scalars, coefficients, comparison maps, or χ-matrices from an MPDO tensor. Once both the same-length BNT product form and a positive BNT-label χ-witness are available, the same-length product expansion with coefficients written as $\mathrm{tr}(\chi^{L}_{\alpha,\beta,\gamma})$ is formalized as a direct corollary.[8] Once both the comparison predicate and a positive BNT-label χ-witness are available, the coefficient-level blocked trace-power formula is formalized as a direct corollary.[9] The same hypotheses also give a positive blocked-basis χ-witness by pulling back the uniform BNT-label χ-family along the comparison maps.[10] Likewise, once the idempotent coefficient condition and the positive BNT-label χ-witness are available, the length-one idempotent identity is formalized with the coefficients rewritten as traces of the corresponding χ-matrices.[11]

The BNT-label objects above are now also gathered into one theorem-data record: fixed coefficients, same-length operators and product law, trace scalars and idempotent condition, positive BNT-label χ-witness, and blocked-basis comparison.[12] This record is not an additional mathematical assumption in the source theorem; it is the single formal object collecting the source data supplied by the Appendix C.3–C.4 construction from an MPDO tensor. There is also a construction for the source-side special case in which the coefficient family is the one canonically determined by the same χ-family; it takes the product law, idempotent law, and blocked-basis comparison for those canonical coefficients.[13] From such data, the source predicates themselves and the corresponding equation-level consequences are available: the same-length product law, the idempotent scalar law, the BNT-label χ-positivity and trace-power predicates, the coefficient trace-power formula, positivity of the diagonal χ-entries, the same-length chi-trace product law, the chi-trace idempotent law, the blocked-basis coefficient trace-power formula, the blocked-basis coefficient comparison, the positive blocked χ-witness, the pulled-back blocked trace-power predicate, and the pulled-back blocked-basis χ-family with positive entries and trace-power formula are immediate consequences.[14]

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[8] MPOTensor.BNTLabelOperatorFamily.HasSameLengthProductForm.eq˙sum˙chi˙trace˙pow, MPOTensor.BNTLabelOperatorFamily.HasSameLengthProductForm.eq˙sum˙ofChi˙trace˙pow.

[9] MPOTensor.BNTBlockedBasisCoefficientComparison.blocked˙coeff˙eq˙trace˙pow, MPOTensor.BNTBlockedBasisCoefficientComparison.blocked˙coeff˙eq˙ofChi˙trace˙pow, MPOTensor.BNTBlockedBasisCoefficientComparison.blocked˙coeff˙eq˙pulledBlockedChi˙trace˙pow.

[10] MPOTensor.BNTBlockedBasisCoefficientComparison.pulledBlockedChiFamily˙toDiagonal˙of˙pos, MPOTensor.BNTBlockedBasisCoefficientComparison.pulledBlockedChi˙tracePowerCoeff˙of˙pos, MPOTensor.BNTBlockedBasisCoefficientComparison.pulledBlockedChi˙dim˙of˙pos, MPOTensor.BNTBlockedBasisCoefficientComparison.pulledBlockedChi˙trace˙matrix˙pow˙of˙pos, MPOTensor.BNTBlockedBasisCoefficientComparison.toPositiveBlockedStructureChiTracePowerForm.

[11] MPOTensor.BNTLabelTraceScalarFamily.HasIdempotentCoefficientForm.eq˙sum˙chi˙trace, MPOTensor.BNTLabelTraceScalarFamily.HasIdempotentCoefficientForm.eq˙sum˙ofChi˙trace.

[12] MPOTensor.BNTLabelTheoremData.

[13] MPOTensor.BNTLabelTheoremData.ofChi.

[14] MPOTensor.BNTLabelTheoremData.same˙length˙product˙form, MPOTensor.BNTLabelTheoremData.idempotent˙coefficient˙form, MPOTensor.BNTLabelTheoremData.positive˙chi˙pos˙entries, MPOTensor.BNTLabelTheoremData.positive˙chi˙trace˙power, MPOTensor.BNTLabelTheoremData.labelAssignment, MPOTen-

The same data can be stated in the existential form closest to the paper: one object records the BNT-label type, the same-length operator spaces, their algebraic structure, and the corresponding theorem data.[15] The existential nonempty statement can also be unpacked directly into a concrete witness carrying the source predicates, the corresponding same-length and idempotent equations written with $c^{(L)}_{\alpha,\beta,\gamma}$, and the same two equations with coefficients written as χ-traces; the blocked-basis χ-family is also identified as the pullback of the BNT-label χ-family along the comparison maps.[16] The existential statement has the analogous canonical construction from a positive χ-family, again leaving the product, idempotent, and blocked-comparison statements as inputs rather

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sor.BNTLabelTheoremData.sourceLabel, MPOTensor.BNTLabelTheoremData.targetLabel, MPOTensor.BNTLabelTheoremData.same˙length˙product˙eq˙sum, MPOTensor.BNTLabelTheoremData.idempotent˙eq˙sum, MPOTensor.BNTLabelTheoremData.coeff˙eq˙trace˙pow, MPOTensor.BNTLabelTheoremData.coeff˙eq˙ofChi˙coeff, MPOTensor.BNTLabelTheoremData.same˙length˙product˙form˙ofChi, MPOTensor.BNTLabelTheoremData.idempotent˙coefficient˙form˙ofChi, MPOTensor.BNTLabelTheoremData.same˙length˙product˙eq˙sum˙ofChi, MPOTensor.BNTLabelTheoremData.idempotent˙eq˙sum˙ofChi, MPOTensor.BNTLabelTheoremData.chi˙entry˙pos, MPOTensor.BNTLabelTheoremData.blocked˙coeff˙eq, MPOTensor.BNTLabelTheoremData.blockedComparison˙ofChi, MPOTensor.BNTLabelTheoremData.blocked˙coeff˙eq˙ofChi, MPOTensor.BNTLabelTheoremData.same˙length˙product˙eq˙sum˙chi˙trace˙pow, MPOTensor.BNTLabelTheoremData.idempotent˙eq˙sum˙chi˙trace, MPOTensor.BNTLabelTheoremData.blocked˙coeff˙eq˙trace˙pow, MPOTensor.BNTLabelTheoremData.toPositiveBlockedStructureChiTracePowerForm, MPOTensor.BNTLabelTheoremData.positiveBlockedChi, MPOTensor.BNTLabelTheoremData.positiveBlockedChi˙toDiagonal˙of˙pos, MPOTensor.BNTLabelTheoremData.positiveBlockedChi˙tracePowerCoeff˙of˙pos, MPOTensor.BNTLabelTheoremData.positiveBlockedChi˙dim˙of˙pos, MPOTensor.BNTLabelTheoremData.positiveBlockedChi˙trace˙matrix˙pow˙of˙pos, MPOTensor.BNTLabelTheoremData.positiveBlockedChi˙tracePower, MPOTensor.BNTLabelTheoremData.positiveBlockedChi˙posEntries, MPOTensor.BNTLabelTheoremData.positiveBlockedChi˙entry˙pos, MPOTensor.BNTLabelTheoremData.blocked˙coeff˙eq˙positiveBlockedChi˙trace˙pow.

[15] MPOTensor.BNTLabelTheoremWitness, MPOTensor.HasBNTLabelTheoremWitness, MPOTensor.BNTLabelTheoremWitness.toTheoremData, MPOTensor.BNTLabelTheoremWitness.coeffs, MPOTensor.BNTLabelTheoremWitness.operators, MPOTensor.BNTLabelTheoremWitness.traceScalars, MPOTensor.BNTLabelTheoremWitness.positiveChi, MPOTensor.BNTLabelTheoremWitness.blockedComparison, MPOTensor.BNTLabelTheoremWitness.same˙length˙product˙form, MPOTensor.BNTLabelTheoremWitness.idempotent˙coefficient˙form, MPOTensor.BNTLabelTheoremWitness.positive˙chi˙pos˙entries, MPOTensor.BNTLabelTheoremWitness.positive˙chi˙trace˙power, MPOTensor.BNTLabelTheoremWitness.labelAssignment, MPOTensor.BNTLabelTheoremWitness.sourceLabel, MPOTensor.BNTLabelTheoremWitness.targetLabel, MPOTensor.BNTLabelTheoremWitness.positiveBlockedChi.

[16] MPOTensor.HasBNTLabelTheoremWitness.exists˙source˙predicates, MPOTensor.HasBNTLabelTheoremWitness.exists˙positive˙length˙coeff˙eq˙ofChi, MPOTensor.HasBNTLabelTheoremWitness.exists˙source˙ofChi˙predicates, MPOTensor.HasBNTLabelTheoremWitness.exists˙source˙ofChi˙equations, MPOTensor.HasBNTLabelTheoremWitness.exists˙blocked˙chi˙pullback, MPOTensor.HasBNTLabelTheoremWitness.exists˙blocked˙coefficient˙comparison, MPOTensor.HasBNTLabelTheoremWitness.exists˙blocked˙coefficient˙comparison˙ofChi, MPOTensor.HasBNTLabelTheoremWitness.exists˙source˙coefficient˙equations, MPOTensor.HasBNTLabelTheoremWitness.exists˙source˙chi˙trace˙equations.

than reconstructing them from an MPDO tensor.[17] Such a witness immediately gives the source predicates and their consequences: the coefficient trace-power formula, the same-length product law, the idempotent scalar law, positivity of the diagonal χ-entries, the blocked-basis coefficient comparison, the same-length chi-trace product law, the chi-trace idempotent law, the blocked-basis coefficient trace-power formula, the positive blocked-basis χ-witness, and the pulled-back blocked trace-power predicate and χ-family for the chosen algebra-structure data.[18] The remaining step is still the construction of this existential witness from the MPDO tensor.

To eliminate the remaining gap, the formalization should introduce:

1. construction of the comparison maps relating the BNT-label coefficients to the chosen blocked bases of $\mathcal{A}_n$;

2. construction of the same-length BNT operator family and product law from the MPDO tensor, together with the comparison to the blocked support-algebra product $\mathcal{A}_n \times \mathcal{A}_n \to \mathcal{A}_{2n}$;

3. construction of the positive BNT-label $\chi_{\alpha,\beta,\gamma}$-witness from the paper's Appendix C.3 and C.4 hypotheses;

[17] MPOTensor.BNTLabelTheoremWitness.ofChi, MPOTensor.BNTLabelTheoremWitness.hasBNTLabelTheoremWitness, MPOTensor.BNTLabelTheoremWitness.ofChi˙hasBNTLabelTheoremWitness.

[18] MPOTensor.BNTLabelTheoremWitness.same˙length˙product˙form, MPOTensor.BNTLabelTheoremWitness.idempotent˙coefficient˙form, MPOTensor.BNTLabelTheoremWitness.positive˙chi˙pos˙entries, MPOTensor.BNTLabelTheoremWitness.positive˙chi˙trace˙power, MPOTensor.BNTLabelTheoremWitness.sourceLabel, MPOTensor.BNTLabelTheoremWitness.targetLabel, MPOTensor.BNTLabelTheoremWitness.coeff˙eq˙trace˙pow, MPOTensor.BNTLabelTheoremWitness.coeff˙eq˙ofChi˙coeff, MPOTensor.BNTLabelTheoremWitness.same˙length˙product˙form˙ofChi, MPOTensor.BNTLabelTheoremWitness.idempotent˙coefficient˙form˙ofChi, MPOTensor.BNTLabelTheoremWitness.same˙length˙product˙eq˙sum˙ofChi, MPOTensor.BNTLabelTheoremWitness.idempotent˙eq˙sum˙ofChi, MPOTensor.BNTLabelTheoremWitness.same˙length˙product˙eq˙sum, MPOTensor.BNTLabelTheoremWitness.idempotent˙eq˙sum, MPOTensor.BNTLabelTheoremWitness.chi˙entry˙pos, MPOTensor.BNTLabelTheoremWitness.blocked˙coeff˙eq, MPOTensor.BNTLabelTheoremWitness.blockedComparison˙ofChi, MPOTensor.BNTLabelTheoremWitness.blocked˙coeff˙eq˙ofChi, MPOTensor.BNTLabelTheoremWitness.same˙length˙product˙eq˙sum˙chi˙trace˙pow, MPOTensor.BNTLabelTheoremWitness.idempotent˙eq˙sum˙chi˙trace, MPOTensor.BNTLabelTheoremWitness.blocked˙coeff˙eq˙trace˙pow, MPOTensor.BNTLabelTheoremWitness.toPositiveBlockedStructureChiTracePowerForm, MPOTensor.BNTLabelTheoremWitness.positiveBlockedChi˙toDiagonal˙of˙pos, MPOTensor.BNTLabelTheoremWitness.positiveBlockedChi˙tracePowerCoeff˙of˙pos, MPOTensor.BNTLabelTheoremWitness.positiveBlockedChi˙dim˙of˙pos, MPOTensor.BNTLabelTheoremWitness.positiveBlockedChi˙trace˙matrix˙pow˙of˙pos, MPOTensor.HasBNTLabelTheoremWitness.positive˙blocked˙chi˙witness, MPOTensor.HasBNTLabelTheoremWitness.exists˙blocked˙chi˙trace˙power˙form, MPOTensor.HasBNTLabelTheoremWitness.exists˙blocked˙coeff˙eq˙trace˙pow, MPOTensor.BNTLabelTheoremWitness.positiveBlockedChi˙tracePower, MPOTensor.BNTLabelTheoremWitness.positiveBlockedChi˙posEntries, MPOTensor.BNTLabelTheoremWitness.positiveBlockedChi˙entry˙pos, MPOTensor.BNTLabelTheoremWitness.blocked˙coeff˙eq˙positiveBlockedChi˙trace˙pow.

4. construction of the idempotent condition $m_\gamma = \sum_{\alpha,\beta} c^{(1)}_{\alpha,\beta,\gamma} m_\alpha m_\beta$ from the MPDO tensor;

5. construction of the single MPOTensor.BNTLabelTheoremWitness from these source objects, and hence of the underlying MPOTensor.BNTLabelTheoremData record. The blocked-basis trace-power statement, with its source-length exponent convention, is then a derived corollary of the uniform BNT-label theorem, rather than a separate replacement for the theorem itself.

References

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